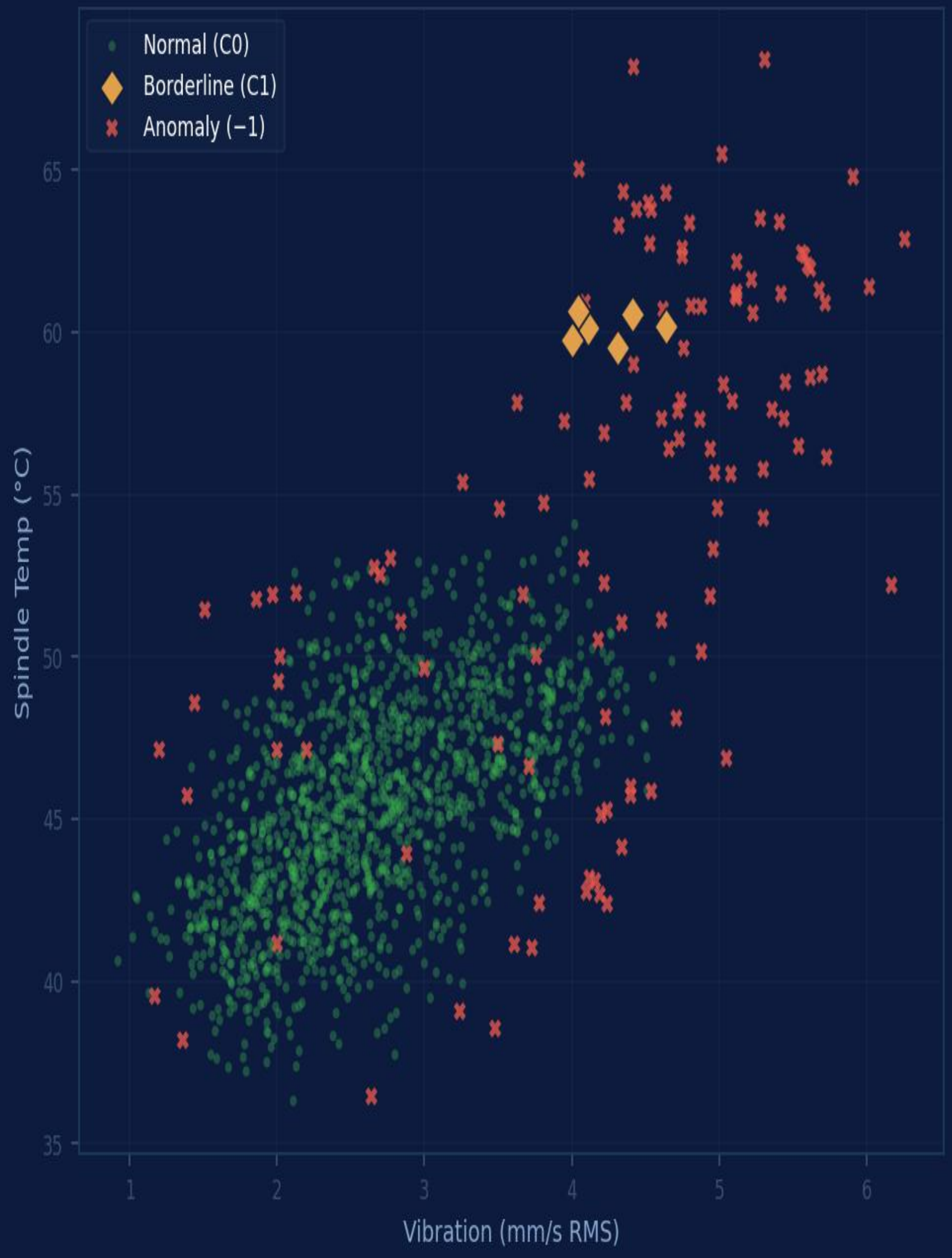
DBSCAN
Anomaly Detection

CNC Machining Operations

*"Does this cycle belong?
Or doesn't it."*

DBSCAN · eps=0.45 | StandardScaler | K-Distance Plot | Density Envelope



1,724
Machining Cycles

149
Anomalies (8.6%)

0.45
eps — Density Radius

1,513
Core Points

The Real Problem

The Threshold Trap

Limits were set years ago when the process was different. They do not update as tools wear, fixtures shift, or cutting conditions evolve. A threshold that made sense at commissioning may be dangerously wide by year three.

Drift, Not Failure

CNC tools do not fail in a moment. They drift. Vibration climbs 0.1 mm/s per shift. Temperature rises 1°C per hundred cycles. Each step looks normal individually. Across all three signals together, the trajectory is unmistakable.

8% of Cycles — 100% of Scrap Risk

149 anomalous cycles: vibration +42%, spindle temp +16%, dimensional deviation +39%. Not one of those cycles triggered a hard alarm. All of them had a distinct density signature that DBSCAN caught.

DBSCAN does not ask whether a reading crossed a line. It asks whether a cycle belongs to the normal population.

A worn tool at 3.80 mm/s passes every threshold. It fails the density test.

From Thresholds to Density

THRESHOLD BY THRESHOLD

IF vibration > 5.0 mm/s → ALARM

IF spindle temp > 65°C → ALARM

IF deviation > 15 µm → ALARM

→ *Each limit set independently*

→ *Variables never interact*

→ *Tool at 3.80 mm/s + 54°C passes*

→ *149 anomalous cycles go undetected*

→ *Worn tool produces scrap silently*

→

DENSITY ENVELOPE

All 3 signals evaluated as a 3D point

eps = 0.45 defines neighborhood radius

min_samples = 5 sets density threshold

→ *Core points form the normal envelope*

→ *Combination pattern is the signal*

→ *Tool at 3.80 + 54°C + 11.6 µm fails*

→ *149 cycles flagged before scrap occurs*

→ *Borderline cases get early warning*

The shift from alarm-based control to density-envelope anomaly detection.

The Data Behind the Problem

SCADA

CNC Controller / Accelerometer

vibration_mm_s

Piezoelectric accelerometer at spindle housing — RMS amplitude computed over full cycle duration

DCS

Machine Tool Controller / RTD

spindle_temp_c

Resistance temperature detector at spindle bearing — logged at cycle end, captures peak thermal load

CMM

Coordinate Measuring Machine / SPC

dim_deviation_um

Post-machining dimensional inspection — deviation of finished part from nominal in micrometers

eps

K-Distance Plot — Hyperparameter Calibration

eps = 0.45 · min_samples = 5

Elbow of the 4th-neighbor distance curve sets the density boundary. Sensitivity: ±0.05 → 6.6%–10.2% anomaly rate

No target variable. DBSCAN maps the density of historical data — the normal envelope emerges from the process itself.

Why DBSCAN?

"The question is not whether any reading crossed a line. It is whether this cycle belongs to the normal population — or not."

No k Required

K-Means needs to be told how many clusters exist. DBSCAN only needs to know how dense normal is. In a CNC cell, you do not know in advance how many failure modes will emerge — DBSCAN does not ask.

Boundary, Not Centroids

The output is not a centroid to describe — it is a frontier. A cycle is either inside the dense normal region or outside it. Label −1 is a binary operational signal, not a category to interpret.

Handles Irregular Shapes

The normal operating envelope of a real process is not spherical. As temperature rises with load, the envelope elongates in that dimension. DBSCAN follows the actual geometry — it imposes no shape assumptions.

−1 Is the Diagnosis

In K-Means every point belongs to some group. In DBSCAN, label −1 is information: this cycle has no neighborhood. It does not fit anywhere in the historical distribution. That is exactly what anomaly detection needs to say.

Before you can flag an anomaly, you need to know the shape of normal.

Model Results

Anomaly Fingerprint

Signal	Normal (C0)	Anomaly (−1)	Delta
Vibration (mm/s)	2.68	3.80	+42%
Spindle Temp (°C)	45.4	52.7	+16%
Dim. Deviation (µm)	8.12	11.32	+39%

No single variable alone exceeded its individual control limit across all 149 anomalous cycles.

Detection Composition

Normal Operation (C0) 1,570

91.1% of all cycles

Borderline Cases (C1) 6

0.4% of all cycles

Anomalies Detected (−1) 149

8.6% of all cycles

eps stability check: ±0.05 → anomaly rate 6.6%–10.2%
K-distance elbow at 0.45 confirmed by sensitivity analysis.

eps = 0.45 · min_samples = 5 · 149 anomalies (8.6%) · 1,513 core points

From Insight to Action

C0

Normal Operation

Trigger: Within envelope · All 3 signals in dense core · Distance to core < eps

→ Continue production. Log centroid values for SPC baseline. No corrective action required.

C1

Borderline — Early Warning

Trigger: Micro-cluster at envelope periphery · Density borderline · 5 cycles (0.3%)

→ Yellow flag. Inspect tool condition at next scheduled break. Measure dimensional output trend. Do not stop production — monitor closely.

−1

Anomaly — Outside Density Envelope

Trigger: Isolated from all core neighborhoods · Vibration > 3.8 mm/s + Temp > 51°C + Dev > 10.5 µm

→ Flag cycle for inspection. Measure finished part against tolerance. Inspect cutting tool for wear or chipping. Tool change probable — escalate to process engineer.

RCA

Root Cause Loop — After Cluster of Anomalies

Trigger: 3+ consecutive anomalous cycles · Or anomaly cluster in production window

→ Correlate with tool change log and maintenance records. Identify which tool life stage the anomalies correspond to. Adjust preventive replacement interval accordingly.

DBSCAN labels are binary — belong or not. The response is graduated.

Batch Classifier — Tool Wear Progression

A — Fresh Tool

Vibration	2.1 mm/s
Spindle Temp	42°C
Dim. Deviation	7.2 µm
Label	C0 — Normal
Neighbors (k=4)	0.28 (< eps)
Status	In dense core

✅ **Normal Operation**

Anomaly Index: 4%

B — Tool at Wear Limit

Vibration	4.5 mm/s
Spindle Temp	55°C
Dim. Deviation	12.1 µm
Label	−1 — Anomaly
Neighbors (k=4)	0.68 (> eps)
Status	Outside envelope

🚨 **Anomaly (−1)**

Anomaly Index: 97%

C — Early Warning

Vibration	**3.4 mm/s**
Spindle Temp	49°C
Dim. Deviation	9.8 µm
Label	C1 — Borderline
Neighbors (k=4)	0.46 (≈ eps)
Status	Peripheral cluster

⚠ **Borderline**

Anomaly Index: 52%

Scenario C — same tool as B, but 20 hours earlier. No threshold has been crossed. Vibration: 3.4 (limit: 5.0). Temp: 49°C (limit: 65°C). DBSCAN already flagged it: borderline. That is the early warning threshold alarms can never provide.

Three moments in the life of a cutting tool — one model, three different answers.

The tool doesn't announce when it's about to fail.

But the density of the process does — if you know how to read it.

Three signals. One envelope. 149 anomalies caught before they became scrap.

> Download the notebook. Run it on your own machining historian data.
>
> *Tune eps. Add features. Add a fourth signal.*
>
> **Because the boundary of normal is never where you think it is.**